

Om Sanjay Singhan

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EDUCATION

New York University, New York, NY— *Dean's List*

September 2023 - May 2025

Master of Science (MS), **Computer Engineering**

GPA: 3.89/4

Coursework: Data Structures & Algorithms, Advanced Machine Learning, Deep Learning, Computational Data Science, Artificial Intelligence, Network Security, Big Data, Computer Systems Architecture, Real-Time Embedded Systems

Mukesh Patel School of Technology Management and Engineering (NMIMS), Mumbai, India

July 2019 - May 2023

Bachelor of Technology in **Computer Engineering**

GPA: 3.76/4

Coursework: Database Management, Machine Learning, Object Oriented Programming, Analysis of Algorithms, Operating Systems

SKILLS

- **Programming Languages:** Java, TypeScript, C#, JavaScript, Python, C/C++, Bash, R, Linux, SQL
- **Backend Frameworks:** GraphQL, ASP .NET, Node.js, Fastify, Express, PostgreSQL, MySQL, NoSQL, MongoDB, DynamoDB
- **Frontend Frameworks:** React, Redux, HTML5, CSS, Tailwind CSS, D3.js, Interactive Dashboards, Data Visualization
- **Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Jenkins, CI/CD, CMake
- **ML Frameworks:** PyTorch, TensorFlow, Keras, NumPy, Pandas, NLTK
- **Agentic AI & LLMs:** GPT-4, LangChain, Agentic Workflows, Prompt Engineering, RAG, MCP Servers, Vector Retrieval.
- **Data & Analytics:** SQL Analytics, Time-Series Data, Tableau, KPI Dashboards, Ad-hoc Analysis, Basic Statistics

PROFESSIONAL EXPERIENCE

Sewer AI, San Francisco, CA

September 2025 – Present

Software Engineer (*Full-time*)

- **Full-Stack API Development:** Engineered secure backend services and RESTful APIs using Node.js, TypeScript, Fastify, and PostgreSQL with Prisma ORM to support data-intensive workflows, handling 10,000+ requests with sub-500ms latency.
- **Test-Driven Development:** Implemented comprehensive test coverage using Jest unit and integration tests and Playwright E2E tests, achieving 85% code coverage and reducing production bugs by 40%.
- **Microservices Architecture:** Architected state-aware, fault-tolerant distributed pipelines using Temporal.io workflow orchestration to process 10,000+ hours of inspection data daily with 99.5% reliability.
- **Cross-Functional Collaboration:** Partnered with product managers and backend engineers to define metrics contracts and ensure trusted, consistent data across APIs and frontend visualizations.

MAK Capital, New York, NY

February 2025 - May 2025

Artificial Intelligence (AI) Developer (*Intern*)

- **Multimodal RAG Integration:** Spearheaded a custom Multimodal RAG pipeline with Azure Document Intelligence, GPT-4, and LangChain, boosting response accuracy by 40% through semantic chunking and hierarchical analysis of text and image data.
- **ETL Pipeline Design:** Architected scalable data ingestion, transformation, and storage pipelines using Python, AWS Lambda, and S3, converting raw JSON into governed datasets for analytics, metrics tracking, and exploratory analysis.
- **Automation & Predictive Modelling:** Built an automated sentiment analysis system with web scraping (BeautifulSoup, Scrapy), FinBERT to identify short opportunities from financial news, delivering 80% prediction accuracy and measurable trading alpha.

Aroris Health, Minneapolis, MN

May 2024 - August 2024

Software Development Engineer (*Intern*)

- **System Design & Data Security:** Improved HIPAA-compliant API infrastructure using Python and Firebase with scalable authentication, protecting 1,000+ patient health records while designing data governance systems for AI-ready pipelines.
- **Product & Data Collaboration:** Collaborated with product managers and UX teams to design and iterate on data centric frontend features for analytics and reporting workflows.
- **Security-First Design:** Implemented OAuth 2.0 authentication, SQL injection prevention, and encrypted data transmission following OWASP guidelines to protect sensitive patient data.

ACADEMIC PROJECTS

NSF Funded Project - Transforming Distribution System Situational Awareness via Continuous-Time Adaptive Data Fusion.

New York University (Generative AI, TimeScaleDB, Express, Python, TensorFlow)

August 2024 - Present

- **AI Simulation & Model Lifecycle:** Designed and trained a power grid simulator using TensorFlow to generate 150,000+ synthetic samples with 88% accuracy, implementing continuous feedback loops for iterative model refinement and drift resilience.
- **Time-Series Analytics & Visualization:** Engineered interactive D3.js dashboards for real-time and historical time-series metrics, applying best practices in chart selection, color hierarchy, and data storytelling while improving query performance by 50%.

Music Recommendation System (Machine Learning, NLP)

July 2022 - May 2023

- **Data Processing:** Employed advanced data science skills to process a dataset of 100k songs with Pandas and NumPy, cutting down processing time by 40% and enhancing accuracy for trend analysis and statistical modeling.
- **Machine Learning:** Developed a recommendation engine using ALS and collaborative filtering algorithms (user-user, item-item similarity, matrix factorization), achieving a 35% improvement in prediction accuracy by optimizing sparse datasets.

LEADERSHIP EXPERIENCE

- Research Assistant - Research Assistant, NYU: Mentored 2 junior developers on code review practices, and design principles.